

CRACK THE CODE

10-week teaching program and scope sequence

Year 7-8 Technology | data and digital systems | ThinkerShield alarm project

Program field	Current position
Syllabus	NSW Technology 7-8 Syllabus (2023), implementation from 2026
Duration	10 weeks; exact lesson allocation Teacher to confirm
Central outcome	TE4-DIG-02 - uses data and digital systems to code, design and produce projects
Primary course evidence	Authorised archived Classroom classwork and teacher-master Drive folder
Final challenge	Design, produce and evaluate a ThinkerShield alarm or alert that protects something the student cares about
Administrative boundary	Dates, marks, rubric, submission route, equipment access and current risk controls: Teacher to confirm

Document status: source-bound implementation draft for faculty review and current-school confirmation.

1. Program rationale and learning design

Crack the Code develops technological literacy through a coherent progression from control-system concepts to an independently planned alarm project. Students read real sensor data, express algorithms in several representations, code fixed inputs and outputs, test predicted behaviour, debug systematically and evaluate the result against evidence.

Student learning package

- Ten dedicated modules; each contains three named theory sections.
- Ten adjacent knowledge-check questions and one guided written response for every theory section.
- A student-controlled video, purposeful diagram and eight-slide learning presentation for every module sequence.
- Ten autosaving activity studios covering all sixteen authorised practical tasks.
- A consolidated device-local folio with backup, restore and print functions.
- Assessment and teacher pages that mark every unset administrative fact as Teacher to confirm.

Course constraints and source corrections

- Fixed inputs: button pin 7, LDR A4, potentiometer A5. Fixed outputs: LEDs pins 8-13 and buzzer pin 3.
- Historical dates and the earlier 20-mark rubric are not treated as current facts.
- Branching uses the new-syllabus meaning: a conditional instruction selects different actions.
- Website autosave is device-local and does not submit work.
- No practical work is authorised outside teacher-directed classroom conditions.

2. NSW Technology 7-8 outcome alignment

Outcome	Authoritative outcome wording	How this unit provides evidence
TE4-SDP-01	explains relationships between sustainability, design and production	Purposeful design and responsible use of equipment and digital work.
TE4-PDP-01	describes the practices and processes of designers and producers	Design, coding, production, testing and evaluation process.
TE4-MSD-01	explains how materials, systems and components contribute to solutions	Sensors, microcontroller, code, LEDs and buzzer as an interacting system.
TE4-PPM-01	applies processes in the planning, management and production of projects	Plan, produce, test and refine the alarm project.
TE4-DES-01	communicates and evaluates design ideas and solutions	IPO, pseudocode, flowcharts, criteria, test evidence and evaluation.
TE4-SAF-01	selects and safely uses tools, materials, technologies and processes	Teacher-directed board handling, connection and testing routines.
TE4-DIG-01	demonstrates technological literacy to safely interact in digital environments	Named versions, backups, sources and confirmed submission processes.
TE4-DIG-02	uses data and digital systems to code, design and produce projects	Central evidence: data, branching, mapping, arrays, code and final alarm.

Source: NESA Technology 7-8 Syllabus (2023) outcomes. School programming decisions determine final emphasis and reporting judgement.

3. Ten-week scope and sequence

Week	Module	Learning focus	Evidence checkpoint	Outcome focus
1	Systems that sense and respond	IPO; microcontroller, shield and sketch roles; fixed pin map.	Hardware map; control-system detective; 30 checks; 3 responses.	TE4-MSC-01, TE4-DIG-02
2	Make light obey the code	setup/loop; digitalWrite; HIGH/LOW; timing; Getting Flashy and Even Flashier.	Predictions, timing comparison and first debugging record.	TE4-DIG-02, TE4-SAF-01
3	Plan the logic before coding	Algorithms, pseudocode, flowcharts and conditional branching.	Algorithm, dry-run and two-path flowchart.	TE4-DIG-02, TE4-DES-01
4	Read changing data	POT Basics, A Light Dimmer, POT LED Bouncer; analogue data, map and calibration.	Measured A5 range, mapping table and calibration justification.	TE4-DIG-02, TE4-MSC-01
5	Sense the light	LDR Basics and LDR Night Light; thresholds and sensor-response loops.	A4 data set, justified threshold, trigger and recovery evidence.	TE4-DIG-02, TE4-PDP-01
6	Make the code choose	Push Button Basics and Get A Toggle On; pin 7, if/else and Boolean state.	Button-state table, branch trace and toggle state trace.	TE4-DIG-02, TE4-MSC-01
7	Turn numbers into sound	Buzzer Basics, Pitch Changer and Play a Song; frequency, mapping and sequences.	Tested frequencies, A5-to-pitch trace and melody code annotation.	TE4-DIG-02, TE4-SAF-01
8	Combine parts into systems	Electronic Dice, Buzzing Light Meter and LED Magic Sign; arrays, loops, functions.	Challenge IPO, index trace and subsystem test order.	TE4-DIG-02, TE4-PPM-01
9	Design the final alarm	Final Alarm Code; brief, criteria, aligned planning and production.	Brief, criteria, pin map, IPO, pseudocode, code record and tests.	TE4-DES-01, TE4-PPM-01, TE4-DIG-02
10	Debug, improve and explain	Systematic debugging, evidence-based evaluation, backup and hand-in readiness.	Debug log, criterion evaluation, improvements and final backup.	TE4-DES-01, TE4-DIG-01, TE4-DIG-02

4. Detailed teaching and learning sequence

Week 1: Systems that sense and respond

Learning intentions	Suggested sequence	Student evidence	Teacher checkpoints
IPO; microcontroller, shield and sketch roles; fixed pin map.	Launch with the module presentation. Teach three theory sections. Students complete adjacent checks and guided responses, then the linked activity studio. Practical work follows the teacher-confirmed lesson allocation.	Hardware map; control-system detective; 30 checks; 3 responses.	Confirm equipment, supervision and current class controls. Check theory, planning and code against the fixed pin map. Outcome focus: TE4-MSC-01, TE4-DIG-02.

Week 2: Make light obey the code

Learning intentions	Suggested sequence	Student evidence	Teacher checkpoints
setup/loop; digitalWrite; HIGH/LOW; timing; Getting Flashy and Even Flashier.	Launch with the module presentation. Teach three theory sections. Students complete adjacent checks and guided responses, then the linked activity studio. Practical work follows the teacher-confirmed lesson allocation.	Predictions, timing comparison and first debugging record.	Confirm equipment, supervision and current class controls. Check theory, planning and code against the fixed pin map. Outcome focus: TE4-DIG-02, TE4-SAF-01.

Week 3: Plan the logic before coding

Learning intentions	Suggested sequence	Student evidence	Teacher checkpoints
Algorithms, pseudocode, flowcharts and conditional branching.	Launch with the module presentation. Teach three theory sections. Students complete adjacent checks and guided responses, then the linked activity studio. Practical work follows the teacher-confirmed lesson allocation.	Algorithm, dry-run and two-path flowchart.	Confirm equipment, supervision and current class controls. Check theory, planning and code against the fixed pin map. Outcome focus: TE4-DIG-02, TE4-DES-01.

Week 4: Read changing data

Learning intentions	Suggested sequence	Student evidence	Teacher checkpoints
POT Basics, A Light Dimmer, POT LED Bouncer; analogue data, map and calibration.	Launch with the module presentation. Teach three theory sections. Students complete adjacent checks and guided responses, then the linked activity studio. Practical work follows the teacher-confirmed lesson allocation.	Measured A5 range, mapping table and calibration justification.	Confirm equipment, supervision and current class controls. Check theory, planning and code against the fixed pin map. Outcome focus: TE4-DIG-02, TE4-MSC-01.

Week 5: Sense the light

Learning intentions	Suggested sequence	Student evidence	Teacher checkpoints
LDR Basics and LDR Night Light; thresholds and sensor-response loops.	Launch with the module presentation. Teach three theory sections. Students complete adjacent checks and guided responses, then the linked activity studio. Practical work follows the teacher-confirmed lesson allocation.	A4 data set, justified threshold, trigger and recovery evidence.	Confirm equipment, supervision and current class controls. Check theory, planning and code against the fixed pin map. Outcome focus: TE4-DIG-02, TE4-PDP-01.

Week 6: Make the code choose

Learning intentions	Suggested sequence	Student evidence	Teacher checkpoints
Push Button Basics and Get A Toggle On; pin 7, if/else and Boolean state.	Launch with the module presentation. Teach three theory sections. Students complete adjacent checks and guided responses, then the linked activity studio. Practical work follows the teacher-confirmed lesson allocation.	Button-state table, branch trace and toggle state trace.	Confirm equipment, supervision and current class controls. Check theory, planning and code against the fixed pin map. Outcome focus: TE4-DIG-02, TE4-MSD-01.

Week 7: Turn numbers into sound

Learning intentions	Suggested sequence	Student evidence	Teacher checkpoints
Buzzer Basics, Pitch Changer and Play a Song; frequency, mapping and sequences.	Launch with the module presentation. Teach three theory sections. Students complete adjacent checks and guided responses, then the linked activity studio. Practical work follows the teacher-confirmed lesson allocation.	Tested frequencies, A5-to-pitch trace and melody code annotation.	Confirm equipment, supervision and current class controls. Check theory, planning and code against the fixed pin map. Outcome focus: TE4-DIG-02, TE4-SAF-01.

Week 8: Combine parts into systems

Learning intentions	Suggested sequence	Student evidence	Teacher checkpoints
Electronic Dice, Buzzing Light Meter and LED Magic Sign; arrays, loops, functions.	Launch with the module presentation. Teach three theory sections. Students complete adjacent checks and guided responses, then the linked activity studio. Practical work follows the teacher-confirmed lesson allocation.	Challenge IPO, index trace and subsystem test order.	Confirm equipment, supervision and current class controls. Check theory, planning and code against the fixed pin map. Outcome focus: TE4-DIG-02, TE4-PPM-01.

Week 9: Design the final alarm

Learning intentions	Suggested sequence	Student evidence	Teacher checkpoints
Final Alarm Code; brief, criteria, aligned planning and production.	Launch with the module presentation. Teach three theory sections. Students complete adjacent checks and guided responses, then the linked activity studio. Practical work follows the teacher-confirmed lesson allocation.	Brief, criteria, pin map, IPO, pseudocode, code record and tests.	Confirm equipment, supervision and current class controls. Check theory, planning and code against the fixed pin map. Outcome focus: TE4-DES-01, TE4-PPM-01, TE4-DIG-02.

Week 10: Debug, improve and explain

Learning intentions	Suggested sequence	Student evidence	Teacher checkpoints
Systematic debugging, evidence-based evaluation, backup and hand-in readiness.	Launch with the module presentation. Teach three theory sections. Students complete adjacent checks and guided responses, then the linked activity studio. Practical work follows the teacher-confirmed lesson allocation.	Debug log, criterion evaluation, improvements and final backup.	Confirm equipment, supervision and current class controls. Check theory, planning and code against the fixed pin map. Outcome focus: TE4-DES-01, TE4-DIG-01, TE4-DIG-02.

5. Assessment evidence and feedback

The supplied course sources establish a final ThinkerShield alarm challenge but do not establish current-year marks, dates, rubric or submission arrangements. The following evidence structure supports formative feedback and can be mapped to a ratified task once confirmed.

Evidence family	Examples	Feedback focus
Concept knowledge	300 checked questions and 30 guided responses	Accuracy, vocabulary, signal-path reasoning and correction.
Practical pathway	Sixteen coding challenges across ten studios	Prediction, observation, trace, calibration and subsystem testing.
Design communication	Brief, criteria, pin map, IPO, pseudocode and flowchart	Agreement, testability and use of constraints.
Production and testing	Sketch record, test cases, results and debugging log	Reliability, boundary and recovery behaviour, change and retest.
Evaluation	Criterion-based judgement, improvements and reflection	Specific evidence, defensible judgement and feasible next action.

Feedback routine

- Return students to the precise relevant theory section before additional help.
- Use the label Appropriate response example; do not supply a teacher answer to copy.
- Separate practice autosave from formal submission and task status.
- Record current faculty decisions in the ratified task notice.

6. Adjustments, accessibility and extension

Need	Planned support	Extension without changing the brief
Literacy and language	Plain-language response guide, theory-return link, sentence starter and model structure.	Require stronger evidence chains, alternatives or code trace.
Executive function	One module at a time, visible contents, autosave, progress, backup and checkpoints.	Plan a more efficient test sequence and justify priorities.
Vision or reading access	High contrast, keyboard access, text alternatives, open-larger diagrams, responsive and print views.	Improve diagram annotation and accessible explanation.
Coding confidence	Prediction, dry-run tables, subsystem tests and one-change debugging.	Refactor using arrays, functions and bounded iteration.
Practical access	Teacher-confirmed pairing, equipment and timing adjustments.	Analyse a second combined-system challenge.

Adjustments preserve access to the learning intention and are decided in the current school context. Student medical, disability or personal information is not stored in this program or website.

7. Safety, equipment and digital responsibility

Area	Established course boundary	Teacher to confirm before delivery
Hardware	Use the supplied Arduino/ThinkerShield and blue USB cable on a clear stable workspace; code matches the fixed board.	Equipment condition, damaged-item procedure, power, storage and supervision.
Sound output	Buzzer on pin 3; code explicitly starts and stops tones.	Acceptable frequency, duration, volume and testing sequence.
Practical lesson	Students predict, verify, upload, observe and record under teacher direction.	Room-specific risk controls, movement, shared equipment and emergency processes.
Digital files	Named versions and a backup; website storage is device-local practice only.	Drive location, file naming, sharing, collaboration and Classroom turn-in.
Privacy	No student submissions, names, grades or private photographs are embedded.	Any student media collection, consent, access and deletion process.

8. Resources, preparation and source ledger

Core resources

- Crack the Code website and ten module presentations.
- Arduino-compatible microcontroller, ThinkerShield, blue USB cable and teacher-approved computers/software.
- Authorised Classroom instructions, activity book and source code documents.
- Teacher-confirmed storage, backup and submission locations.

Authoritative sources

Source	Use in this program
Archived 2025 Crack the Code Classroom classwork and teacher-master Drive folder	Activity names, order, hardware map, coding pathway and final alarm challenge.
NESA Technology 7-8 Syllabus (2023) overview, outcomes and glossary	Implementation boundary, Stage 4 outcome wording and branching definition.
Teacher confirmation before implementation	Dates, marks, rubric, lessons, equipment, supervision, risk controls and submission.

Review and sign-off

Review item	Name / decision / date
Faculty review	
Current assessment status	Teacher to confirm
Risk and equipment review	Teacher to confirm
Lesson allocation	Teacher to confirm
Submission and storage process	Teacher to confirm